

Visually Descriptive Language Model for Vector Graphics Reasoning

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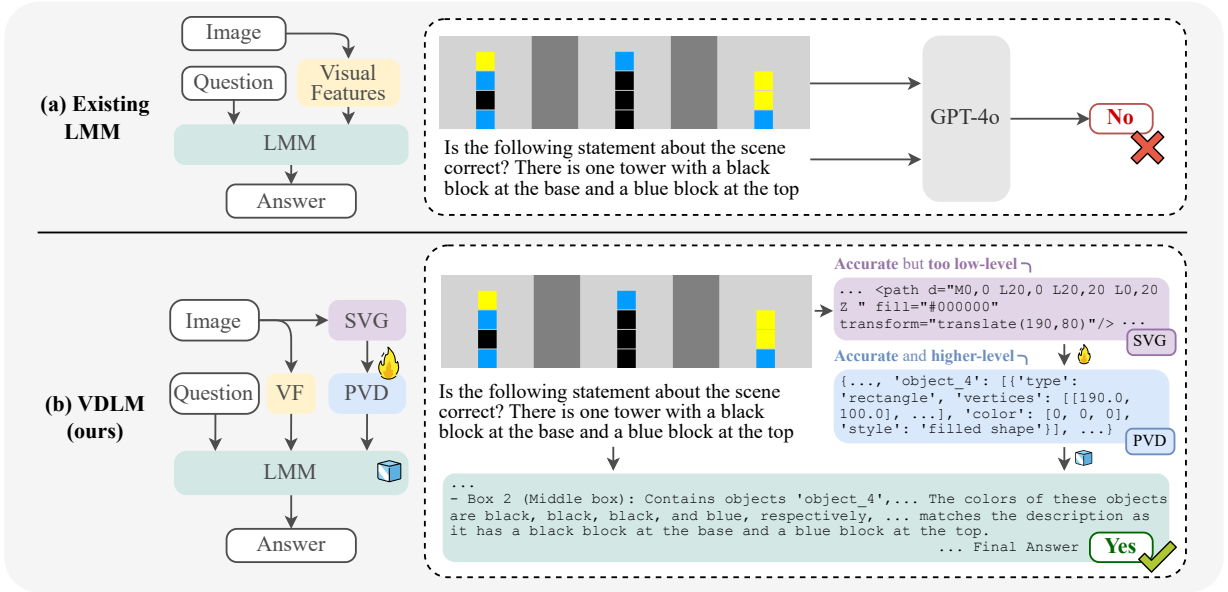


Figure 1: Existing monolithic LMMs rely solely on pretrained vision encoders, such as CLIP (Radford et al., 2021), for perception, which often fail to accurately capture low-level visual details in vector graphics-style images. In contrast, VDLM enables precise visual reasoning by introducing SVG encoding and a learned intermediate symbolic representation, Primal Visual Description (PVD), that bridges low-level SVG perception with high-level language reasoning. “VF” in (b) refers to “visual features”.

Abstract

Despite significant advancements, current large multimodal models (LMMs) struggle to bridge the gap between low-level visual perception—focusing on shapes, sizes, and layouts—and high-level language reasoning involving semantics, events, and logic. This limitation becomes evident in tasks requiring precise visual perception, such as comparing geometric properties or solving visual algorithmic reasoning problems. To study this failure mode, we focus on an important visual domain: vector graphics—images composed purely of 2D objects and shapes, which are prevalent in web and mobile environments. Importantly, we consider *rasterized* vector graphics without assuming access to their underlying vector code. We identify two key research questions: how can we enable precise visual perception, and how can we facilitate high-level reasoning based on such low-level perceptions? To accurately capture low-level visual details, we explore using SVG for the precise encoding of visual scenes.

However, SVGs are not readily interpretable by LLMs or LMMs in a zero-shot manner. To address this challenge, we propose the **Visually Descriptive Language Model (VDLM)** to build a bridge between low-level visual perception and high-level language reasoning. VDLM learns an intermediate symbolic representation called **Primal Visual Description (PVD)**, which translates raw SVGs into a higher-level abstraction comprising primitive attributes. This abstraction allows for *direct interpretation* by foundation models for zero-shot generalization to different reasoning tasks. As an initial step to construct a descriptive intermediate representation for low-level visual reasoning, the SVG-to-PVD model is currently limited to simple compositions of primitive shapes, for which synthetic data can be generated without human annotation. Nevertheless, empirical experiments show that VDLM leads to significant improvements in state-of-the-art LMMs, such as GPT-4o, across various low-level multimodal perception and reasoning tasks on rasterized vector graphics. Additionally, we provide extensive analyses of VDLM’s performance, showing that our framework offers improved interpretability due to its disentangled perception and reasoning processes. We also conduct an in-depth error analysis, highlighting remaining limitations and suggesting directions for future research. Project page: <https://mikewangwzh1.github.io/VDLM/>

1 Introduction

In recent years, large multimodal models (LMMs) (OpenAI, 2023b; Anil et al., 2023; Liu et al., 2023b; Chen et al., 2023b; Bai et al., 2023) have achieved impressive performance across a wide spectrum of general vision-language benchmarks (Goyal et al., 2017; Fu et al., 2023; Liu et al., 2023d; Yu et al., 2023; Li et al., 2023a). However, these monolithic LMMs still struggle with seemingly simple tasks that require precise perception of low-level visual details. In particular, we empirically observe that LMMs frequently exhibit this failure mode in vector graphics, which are images composed purely of 2D objects and shapes. For example, a state-of-the-art LMM like GPT-4o (OpenAI, 2024) can still fail 43% of the time when comparing the lengths of two line segments, and 54% of the time when solving a simple 2×2 maze. LMMs’ ability to understand vector graphics is largely underexplored compared to natural images but is essential for growing downstream applications in web, visual design, and OS environments (Zhou et al., 2023b; Liu et al., 2024; Xie et al., 2024; Rawles et al., 2024; Zheng et al., 2024; Lù et al., 2024). In this work, we focus on the fundamental aspects of low-level visual reasoning involving vector graphics—including measurements, spatial relations, counting, and logical reasoning. It is important to note that in this paper, “vector graphics” refers to **rasterized images** in JPEG or PNG format, without assuming access to their underlying vector code. This reflects a more realistic setting for visual reasoning in real-world scenarios, such as web Zhou et al. (2023b); Lù et al. (2024) and mobile Wang et al. (2024; 2025). To address the aforementioned challenge, we identify two main research questions. First, how can we enable precise visual perception in LMMs? Second, how can we bridge the gap between low-level perception and high-level reasoning?

For our initial question, we explore encoding a rasterized image via vectorization with SVG representation, which describes a scene with paths (e.g., polygons and splines) and their corresponding measurements and positions. SVG representations, by nature, are unbiased towards high-level semantics and can capture low-level visual details in text. The vectorization process can be faithfully accomplished with a rule-based raster-to-vector algorithm. However, such machine-generated SVG is often noisy and far from natural language, making it insufficient for language reasoning. Our preliminary experiments (§A) demonstrate that existing foundation models are unable to interpret machine-generated SVG codes in zero-shot settings. Another key challenge lies in the scarcity of end-to-end instruction tuning data containing ⟨SVG, question, answer⟩ triplets, making direct fine-tuning infeasible.

To bridge this perception-reasoning gap and address data scarcity, we propose translating the low-level SVG paths to a higher-level intermediate symbolic representation, referred to as **Primal Visual Description (PVD)**, which can directly be leveraged by foundation models for multimodal reasoning. Specifically, we learn an LLM-based (Jiang et al., 2023) SVG-to-PVD model, which transforms the raw SVG paths into a set of primitive attributes (e.g., shape, position) with corresponding predicted values (e.g., rectangle, pixel coordinates of the vertices). See Figure 1 in the blue box for an example. Notably, the PVD representation

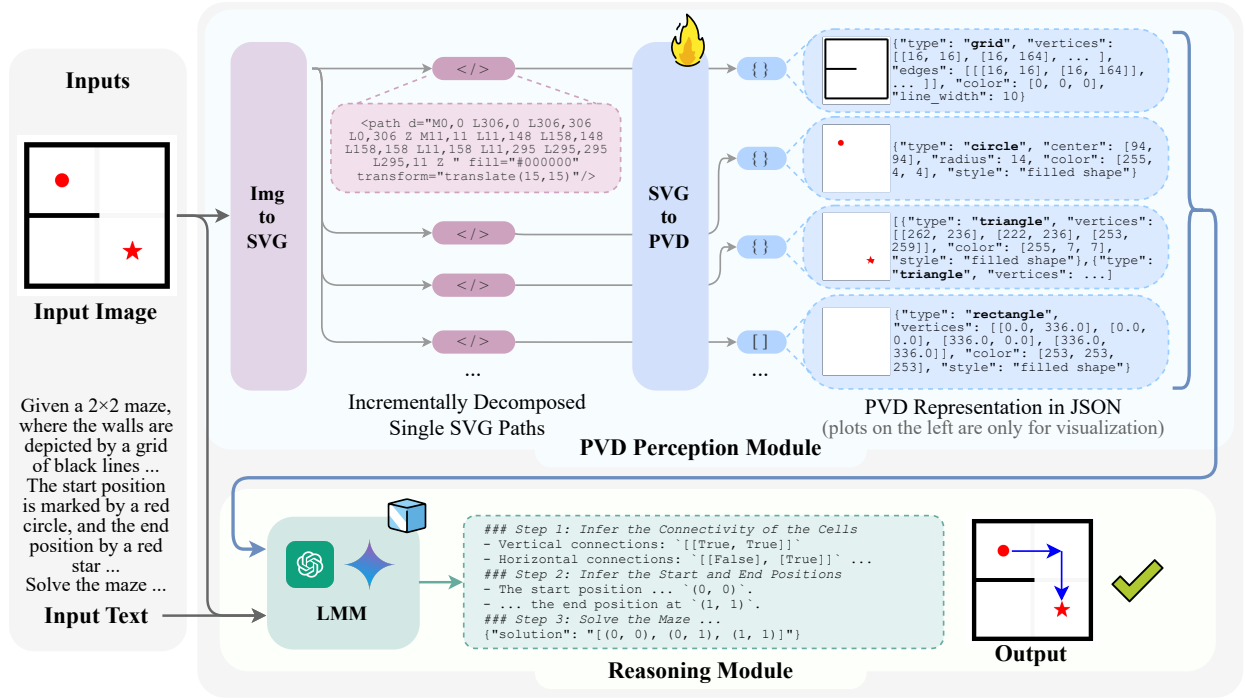


Figure 2: An example of VDLM during inference. First, VDLM extracts individual SVG paths from the input image and then transforms them into Primal Visual Description (PVD) using a trained language model. These PVD perception results, along with the input text queries and the original input image, are subsequently fed into an LMM for reasoning. It is worth noting that although a “star” (★) is not explicitly part of the PVD primitive ontology (see Figure 3), the SVG-to-PVD model can approximate the “star” by composing two triangles (▲). A strong off-the-shelf reasoner, such as GPT-4 (OpenAI, 2023a), can accurately deduce that this composition corresponds to the “star,” which is the target end position of the maze. For the complete response, refer to Figure 17.

consists of primitive attributes that serve as fundamental building blocks for vector graphics, enabling learning from procedurally synthesized $\langle \text{SVG}, \text{PVD} \rangle$ pairs without requiring task-specific annotations. PVD filters out unnecessary noise in SVGs, enhancing the semantics of perception and facilitating subsequent reasoning. Currently, we limit the scope of the SVG-to-PVD model to simple compositions of geometric primitives, for which synthetic data can be generated without human annotation. We leave generalization to entirely open-domain images as future work.

Comprising SVG-based image perception and PVD abstractions, we present the **Visually Descriptive Language Model (VDLM)**, which contains three components: a rule-based visual encoder that converts images to SVG to capture precise visual details, a learned language model that translates SVG to PVD, and an inference-only reasoner that conducts zero-shot reasoning about downstream tasks with the PVD representation. An overview of VDLM is provided in Figure 1. Experimental results demonstrate that VDLM achieves strong zero-shot performance in various visual reasoning tasks, outperforming LLaVA-v1.5 (Liu et al., 2023a), G-LLaVA (Gao et al., 2023), GPT-4V (OpenAI, 2023b), GPT-4o (OpenAI, 2024), and Visual Programming approaches such as ViperGPT (Surís et al., 2023). Moreover, VDLM also enhances interpretability through better disentanglement of perception and reasoning processes.

To summarize, the key contributions of our work are threefold: First, we identify a critical failure mode of LMMs when reasoning about tasks that require precise, low-level perception. Second, we introduce VDLM, the first attempt to enhance LMMs’ fine-grained visual reasoning capabilities with vectorized and symbolic visual descriptions—SVG representations and learned Primal Visual Description. Finally, we present an in-depth analysis of the emergent patterns, the disentangled impact of PVD quality on end-task performance, and the remaining limitations, highlighting directions for future work.